

2. Magnesium oxide helps prevent scale formation in boilers, heat exchangers, and piping. Burning magnesium in a gas jar of oxygen produces magnesium oxide on a small scale.



- (a) The symbol for oxygen is $^{16}_8\text{O}$.

Complete the following statements about an atom of oxygen.

[3]

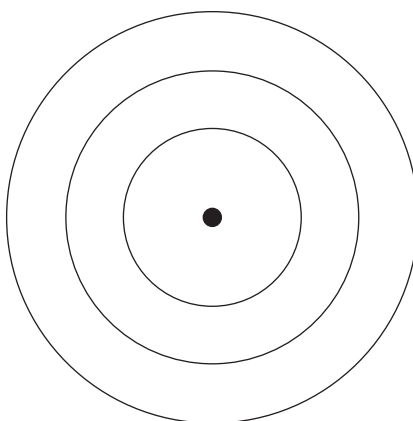
An oxygen atom contains electrons.

An oxygen atom contains neutrons.

An oxygen atom contains protons.

- (b) An atom of magnesium contains 12 electrons. Complete the diagram to show the electronic structure of a magnesium atom.

[1]



- (c) State what happens to a magnesium atom when it becomes a magnesium ion.

[1]

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- (d) **Complete** the balanced symbol equation for the following reaction.

[3]

magnesium + oxygen $\longrightarrow$ magnesium oxide

